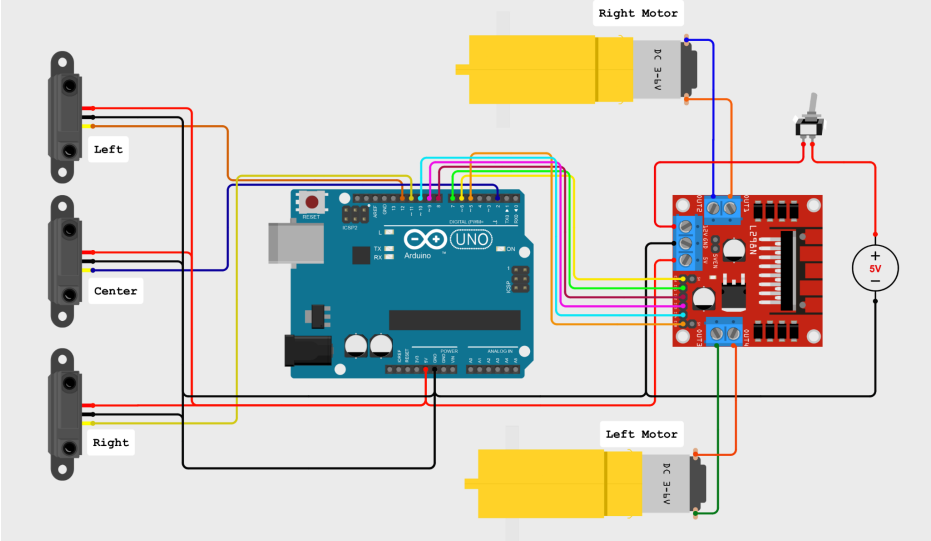
Arduino: Line Follower Robot with 3-IR sensors



Main Components

Material	Quantity	Purpose
Arduino Uno/Nano	1	Main controller
IR line-tracking sensors	3	Detect left, center, and right position of the line
L298N motor driver	1	Controls the two DC motors
TT gear motors	2	Drive the robot
Robot chassis	1	Holds all components
Robot wheels	2	Connected to the motors
Caster wheel	1	Front/rear balance and support
Jumper wires	Several	Electrical connections
Breadboard / mini breadboard	1	Prototyping connections
Battery holder	1	Holds batteries
Batteries	4× AA or suitable pack	Power source
USB cable	1	Arduino programming/power

1. 3-IR sensor configuration

Component	Arduino Uno pin
Left IR sensor OUT	**D12**
Center IR sensor OUT	**D2**
Right IR sensor OUT	**D11**
L298N ENA (right motor PWM)	**D6**
L298N IN1 / IN2	**D7/D8**
L298N ENB (left motor PWM)	**D5**
L298N IN3 / IN4	**D9/10**

Keep the same motor connections shown in your diagram. The added center sensor needs its own signal wire to D2, plus power and ground.

Important: Your IR modules appear to be digital-output sensors. The code below assumes each sensor outputs **LOW** on a black line and **HIGH** on a white surface. Test this first because some modules behave in the opposite way.

2. How to adjust and test the robot

Start with these settings, then tune the robot on the actual track.

Setting	Starting value	Purpose
BASE_SPEED	150	Normal forward movement
TURN_SPEED	85	Slower wheel during gentle turns
SEARCH_SPEED	100	Recover the line after losing it
MAX_SPEED	220	Limits maximum PWM output

Test the sensor readings first

Before running the full track, place the sensors over a black line and a white surface. Confirm their readings:

LEFT	CENTER	RIGHT	EXPECTION ACTION
White	Black	White	Forward
Black	Black	White	Turn left
White	Black	Black	Turn right
Black	White	White	Sharp left
White	White	Black	Sharp right
White	White	White	Search for line

Here, ***black*** means the sensor detects the line, and ***white*** means it detects the background. The table describes typical patterns; intersections and wide lines can produce other combinations.

If your sensors output `HIGH` on black instead, reverse the three sensor comparisons in the code.

3. Important hardware checks

- **Center sensor**: Connect its **OUT** pin to Arduino D2, **VCC** to the appropriate supply for your module, and **GND** to common ground.
- **L298N**: Ensure the **ENA** and **ENB** jumpers are removed if using Arduino *PWM* on *D6* and *D5*.
- **Power**: *Use a suitable motor supply for your motors and driver*, and connect the Arduino and driver grounds together. Avoid powering the motors directly from the Arduino 5V pin.
- **PWM adjustment**: I removed the original **TCCR0B** frequency modification because it affects the Arduino Uno's timer used by ***millis()*** and ***delay()***. Start with the standard PWM frequency and adjust only if testing shows you need to.

My recommendation: First test each motor independently, then test the three sensors, and finally run the robot at a lower speed on a simple oval track. If it turns the wrong way, loses the line on curves, or spins at intersections, the correction will depend on the actual sensor readings and motor orientation.